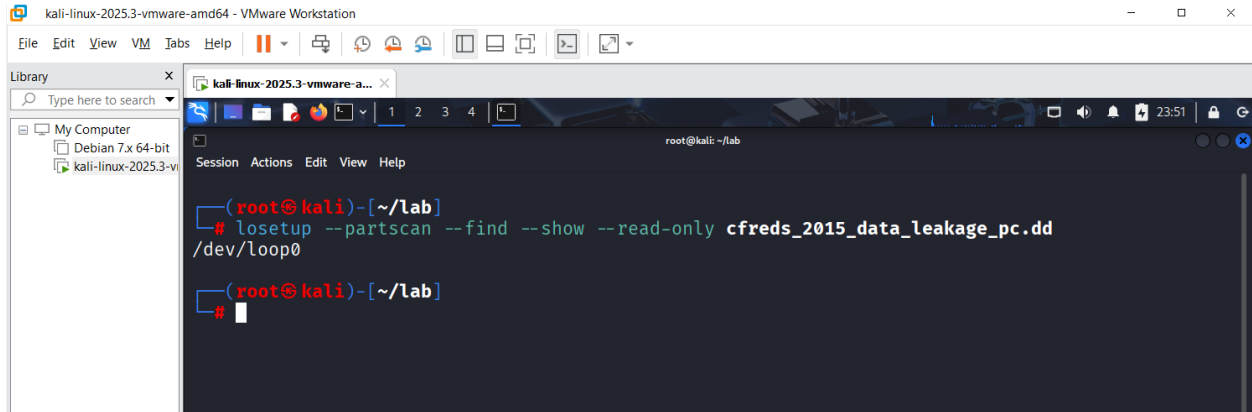


Nguyễn Phạm Trí CE192008

LAB 11

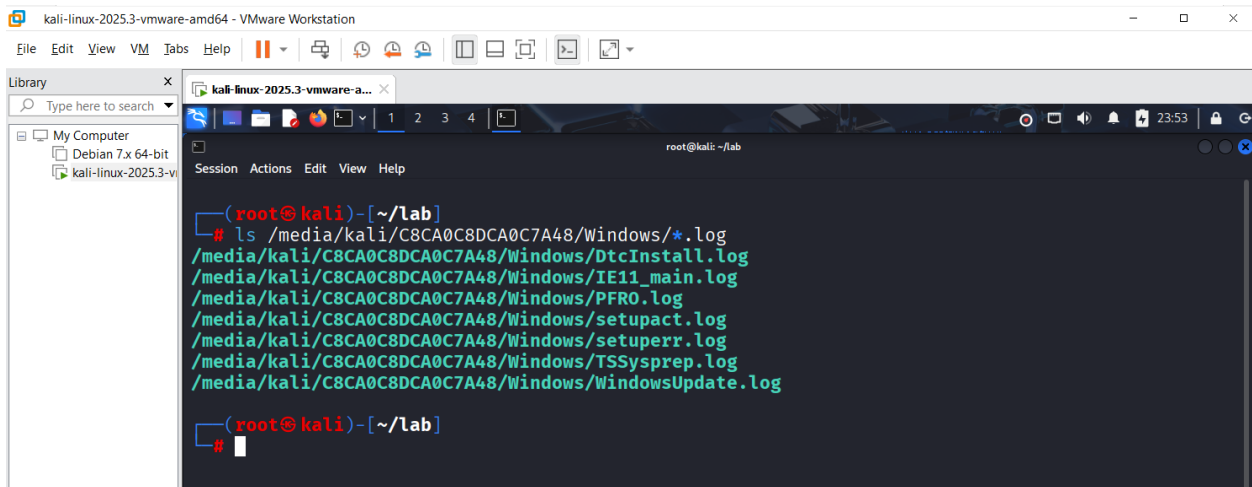
Step 1



The screenshot shows a VMware Workstation window titled "kali-linux-2025.3-vmware-amd64 - VMware Workstation". The terminal window is titled "kali-linux-2025.3-vmware-a..." and shows the following commands and output:

```
(root@kali)-[~/Lab]
# losetup --partscan --find --show --read-only cfreds_2015_data_leakage_pc.dd /dev/loop0
(root@kali)-[~/Lab]
#
```

ảnh trên: thiết lập ảnh đĩa



The screenshot shows a VMware Workstation window titled "kali-linux-2025.3-vmware-amd64 - VMware Workstation". The terminal window is titled "kali-linux-2025.3-vmware-a..." and shows the following commands and output:

```
(root@kali)-[~/Lab]
# ls /media/kali/C8CA0C8DCA0C7A48/Windows/*.log
/media/kali/C8CA0C8DCA0C7A48/Windows/DtcInstall.log
/media/kali/C8CA0C8DCA0C7A48/Windows/IE11_main.log
/media/kali/C8CA0C8DCA0C7A48/Windows/PFR0.log
/media/kali/C8CA0C8DCA0C7A48/Windows/setupact.log
/media/kali/C8CA0C8DCA0C7A48/Windows/setuperr.log
/media/kali/C8CA0C8DCA0C7A48/Windows/TSSysprep.log
/media/kali/C8CA0C8DCA0C7A48/Windows/WindowsUpdate.log
(root@kali)-[~/Lab]
#
```

ảnh trên: Kiểm tra tất cả các phiên bản IE thông qua .log

kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Library

Type here to search

My Computer

Debian 7.x 64-bit

kali-linux-2025.3-vmware-amd64

```
(root@kali)-[~/lab]
# cat /media/kali/C8CA0C8DCA0C7A48/Windows/IE11_main.log | head
00:00.000:
00:00.000: Started: 2015/03/22 (Y/M/D) 11:12:32.577 (local)
00:00.000: Time Format in this log: MM:ss.mmm (minutes:seconds.milliseconds)
00:00.000: Command line: "C:\Users\informant\Desktop\Download\IE11-Windows6.1-x64-en-us.exe"
00:00.000: INFO: Setup installer for Internet Explorer: 11.0.9600.16428
00:00.000: INFO: Previous version of Internet Explorer: 8.0.7601.17514
00:00.015: INFO: Checking if iexplore.exe's current version is between 11.0.9600.0 ...
00:00.015: INFO: ...and 11.1.0.0 ...
00:00.015: INFO: Maximum version on which to run IEAK branding is: 11.1.0.0 ...
00:00.015: INFO: iexplore.exe version check success. Install can proceed.

(root@kali)-[~/lab]
#
```

kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Library

Type here to search

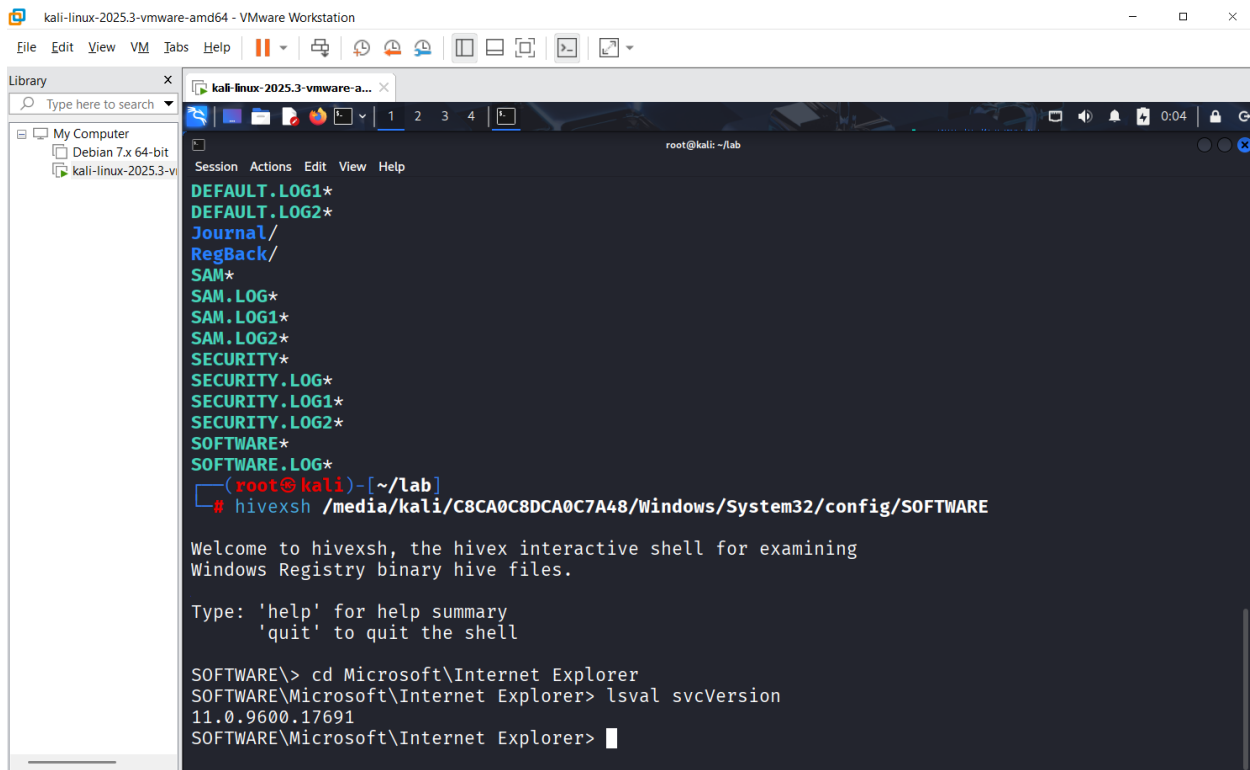
My Computer

Debian 7.x 64-bit

kali-linux-2025.3-vmware-amd64

```
00:00.015: INFO: ...and 11.1.0.0 ...
00:00.015: INFO: Maximum version on which to run IEAK branding is: 11.1.0.0 ...
00:00.015: INFO: iexplore.exe version check success. Install can proceed.

(root@kali)-[~/lab]
# apt-get install libhivex-bin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libplacebo349 librav1e0.7 libtheoradec1 libtheoraenc1 libx264-164
Use 'apt autoremove' to remove them.
The following additional packages will be installed:
  libhivex0
The following NEW packages will be installed:
  libhivex-bin libhivex0
0 upgraded, 2 newly installed, 0 to remove and 614 not upgraded.
Need to get 72.9 kB of archives.
After this operation, 341 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libhivex0 amd64 1.3.24-1+b12 [27.2 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libhivex-bin amd64 1.3.24-1+b12 [45.8 kB]
Fetched 72.9 kB in 1s (68.5 kB/s)
```



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Debian 7.x 64-bit
kali-linux-2025.3-vmware-amd64
kali-linux-2025.3-vmware-amd64
root@kali: ~/lab
Session Actions Edit View Help
DEFAULT.LOG1*
DEFAULT.LOG2*
Journal/
RegBack/
SAM*
SAM.LOG*
SAM.LOG1*
SAM.LOG2*
SECURITY*
SECURITY.LOG*
SECURITY.LOG1*
SECURITY.LOG2*
SOFTWARE*
SOFTWARE.LOG*
(root@kali)~[~/lab]
# hivexsh /media/kali/C8CA0C8DCA0C7A48/Windows/System32/config/SOFTWARE

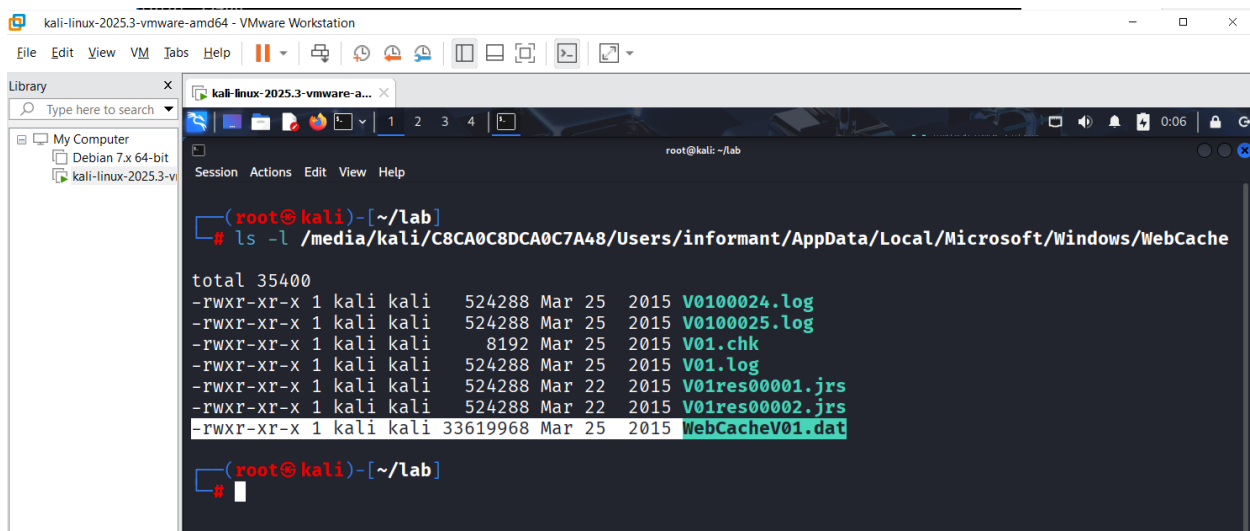
Welcome to hivexsh, the hivex interactive shell for examining
Windows Registry binary hive files.

Type: 'help' for help summary
      'quit' to quit the shell

SOFTWARE\> cd Microsoft\Internet Explorer
SOFTWARE\Microsoft\Internet Explorer> lsval svcVersion
11.0.9600.17691
SOFTWARE\Microsoft\Internet Explorer>
```

ảnh trên: dùng hivexsh (đọc Registry offline) để tìm phiên bản IE (khóa trong hive SOFTWARE). Biết version giúp đoán cấu trúc lịch sử duyệt (IE8 dùng index.dat, IE11 dùng ESE WebCacheV01.dat).

Step 2.

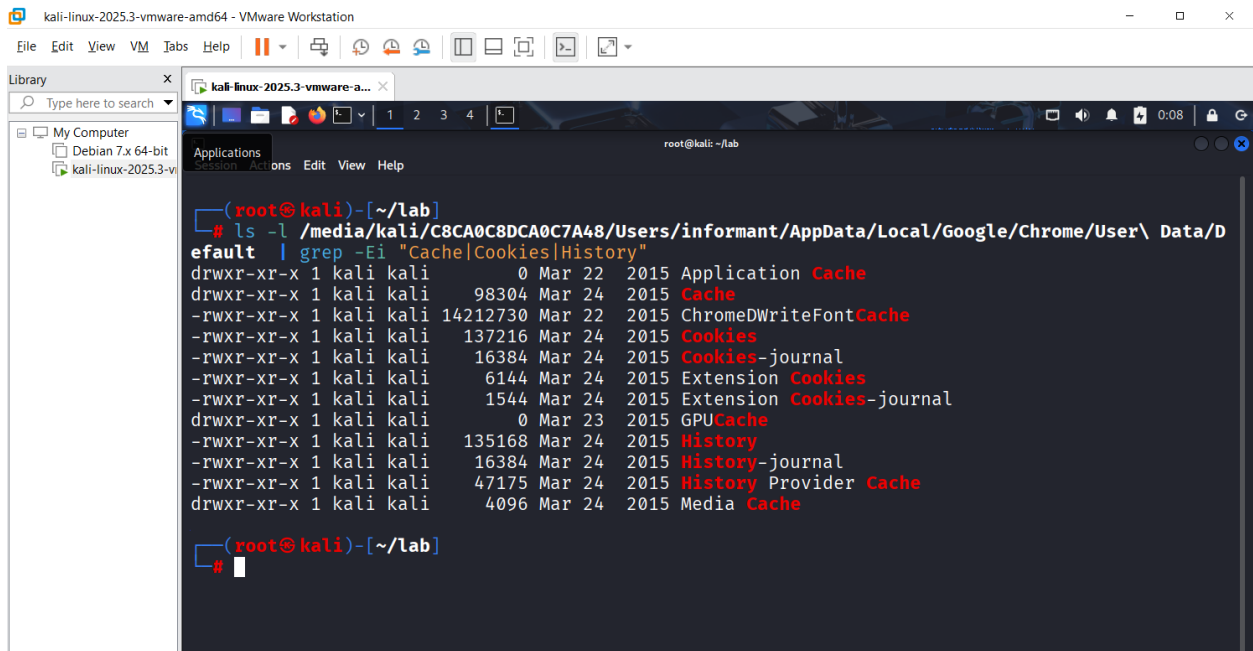


```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Debian 7.x 64-bit
kali-linux-2025.3-vmware-amd64
kali-linux-2025.3-vmware-amd64
root@kali: ~/lab
Session Actions Edit View Help
(root@kali)~[~/lab]
# ls -l /media/kali/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Microsoft/Windows/WebCache

total 35400
-rwxr-xr-x 1 kali kali 524288 Mar 25 2015 V0100024.log
-rwxr-xr-x 1 kali kali 524288 Mar 25 2015 V0100025.log
-rwxr-xr-x 1 kali kali 8192 Mar 25 2015 V01.chk
-rwxr-xr-x 1 kali kali 524288 Mar 25 2015 V01.log
-rwxr-xr-x 1 kali kali 524288 Mar 22 2015 V01res00001.jrs
-rwxr-xr-x 1 kali kali 524288 Mar 22 2015 V01res00002.jrs
-rwxr-xr-x 1 kali kali 33619968 Mar 25 2015 WebCacheV01.dat

(root@kali)~[~/lab]
#
```

ảnh trên: Xác định đường dẫn lịch sử IE11



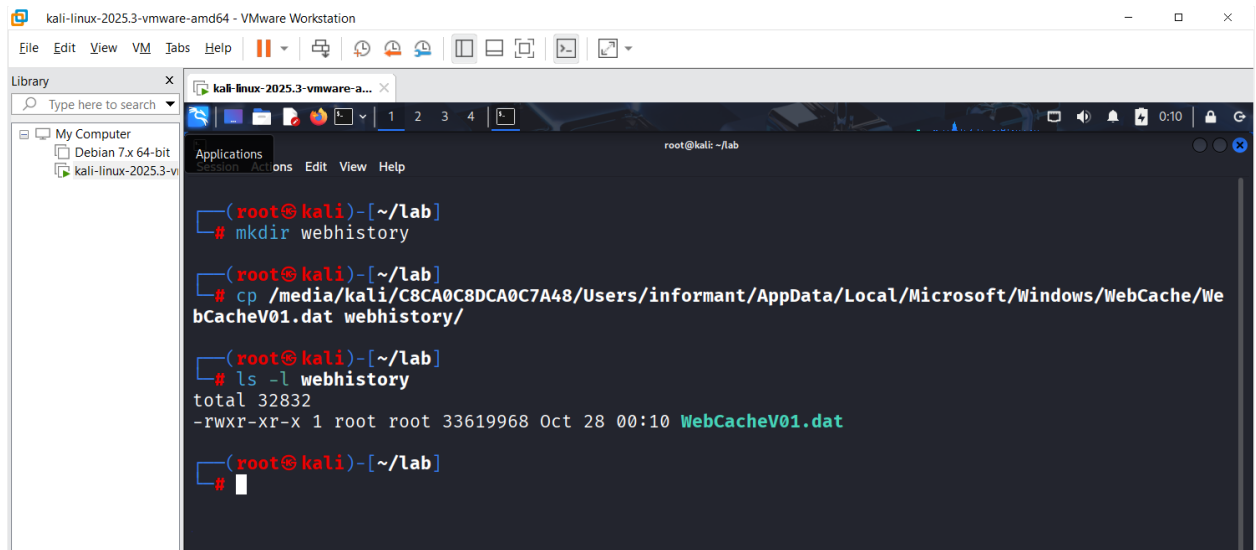
```
(root@kali)-[~/Lab]
# ls -l /media/kali/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Google/Chrome/User\ Data/Default | grep -Ei "Cache|Cookies|History"
drwxr-xr-x 1 kali kali      0 Mar 22  2015 Application Cache
drwxr-xr-x 1 kali kali  98304 Mar 24  2015 Cache
-rwxr-xr-x 1 kali kali 14212730 Mar 22  2015 ChromeDWriteFontCache
-rwxr-xr-x 1 kali kali  137216 Mar 24  2015 Cookies
-rwxr-xr-x 1 kali kali  16384 Mar 24  2015 Cookies-journal
-rwxr-xr-x 1 kali kali   6144 Mar 24  2015 Extension Cookies
-rwxr-xr-x 1 kali kali   1544 Mar 24  2015 Extension Cookies-journal
drwxr-xr-x 1 kali kali      0 Mar 23  2015 GPUCache
-rwxr-xr-x 1 kali kali  135168 Mar 24  2015 History
-rwxr-xr-x 1 kali kali  16384 Mar 24  2015 History-journal
-rwxr-xr-x 1 kali kali  47175 Mar 24  2015 History Provider Cache
drwxr-xr-x 1 kali kali   4096 Mar 24  2015 Media Cache
```

ảnh trên: Xác nhận đường dẫn file lịch sử của Chrome (Cache/Cookies/History).

What websites were the suspect accessing? (Timestamp, URL...)

- Get browser history files
- View these files

Step 1.



```
(root@kali)-[~/Lab]
# mkdir webhistory

(root@kali)-[~/Lab]
# cp /media/kali/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat webhistory/

(root@kali)-[~/Lab]
# ls -l webhistory
total 32832
-rwxr-xr-x 1 root root 33619968 Oct 28 00:10 WebCacheV01.dat
```

ảnh trên: copy file history của IE11

The screenshot shows a terminal window in a VMware Workstation environment. The user is root at a kali machine in the /lab directory. They execute the command `cp /media/kali/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Microsoft/Windows/Temporary\ Internet\ Files/Low/Content.IE5/index.dat webhistory`. Then, they run `ls -l webhistory`, which shows two files: `index.dat` (245760 bytes, Oct 28 00:12) and `WebCacheV01.dat` (33619968 bytes, Oct 28 00:10).

```
root@kali: ~/lab
Session Actions Edit View Help
(root@kali)~[/lab]
# cp /media/kali/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Microsoft/Windows/Temporary\
Internet\ Files/Low/Content.IE5/index.dat webhistory

(root@kali)~[/lab]
# ls -l webhistory
total 33072
-rwxr-xr-x 1 root root 245760 Oct 28 00:12 index.dat
-rwxr-xr-x 1 root root 33619968 Oct 28 00:10 WebCacheV01.dat

(root@kali)~[/lab]
#
```

ảnh trên: Copy IE 8 History

The screenshot shows a terminal window in a VMware Workstation environment. The user is root at a kali machine in the /lab directory. They execute the command `cp /media/kali/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Google/Chrome/User\ Data/Default/ History webhistory`. Then, they run `ls -l webhistory`, which shows three files: `History` (135168 bytes, Oct 28 00:13), `index.dat` (245760 bytes, Oct 28 00:12), and `WebCacheV01.dat` (33619968 bytes, Oct 28 00:10).

```
root@kali: ~/lab
Applications
Session Actions Edit View Help
(root@kali)~[/lab]
# cp /media/kali/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Google/Chrome/User\ Data/Default/
History webhistory

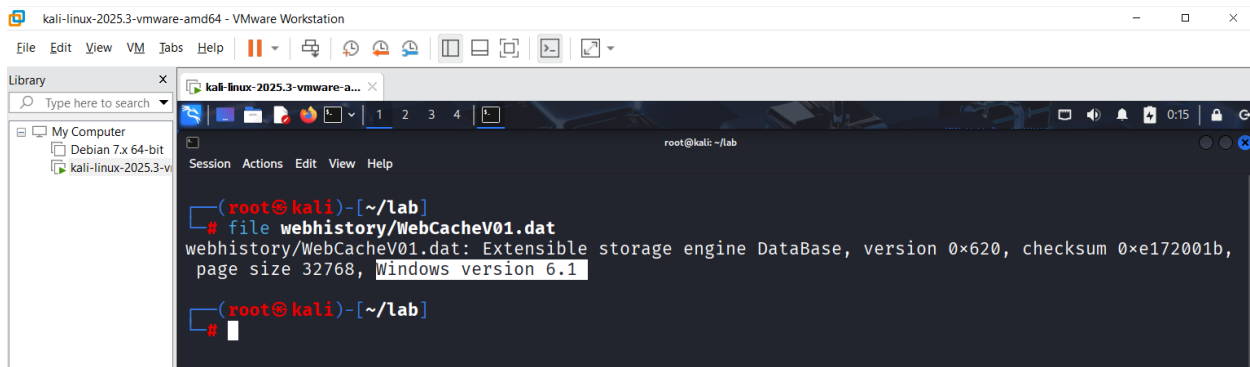
(root@kali)~[/lab]
# ls -l webhistory
total 33204
-rwxr-xr-x 1 root root 135168 Oct 28 00:13 History
-rwxr-xr-x 1 root root 245760 Oct 28 00:12 index.dat
-rwxr-xr-x 1 root root 33619968 Oct 28 00:10 WebCacheV01.dat

(root@kali)~[/lab]
#
```

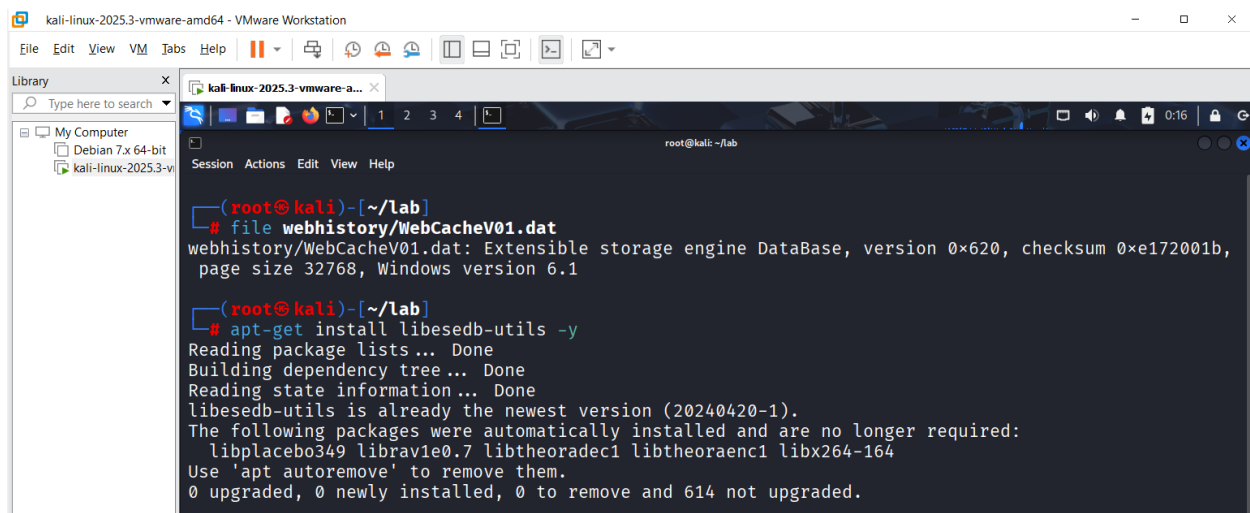
ảnh trên: copy Chrome history

Step 2.

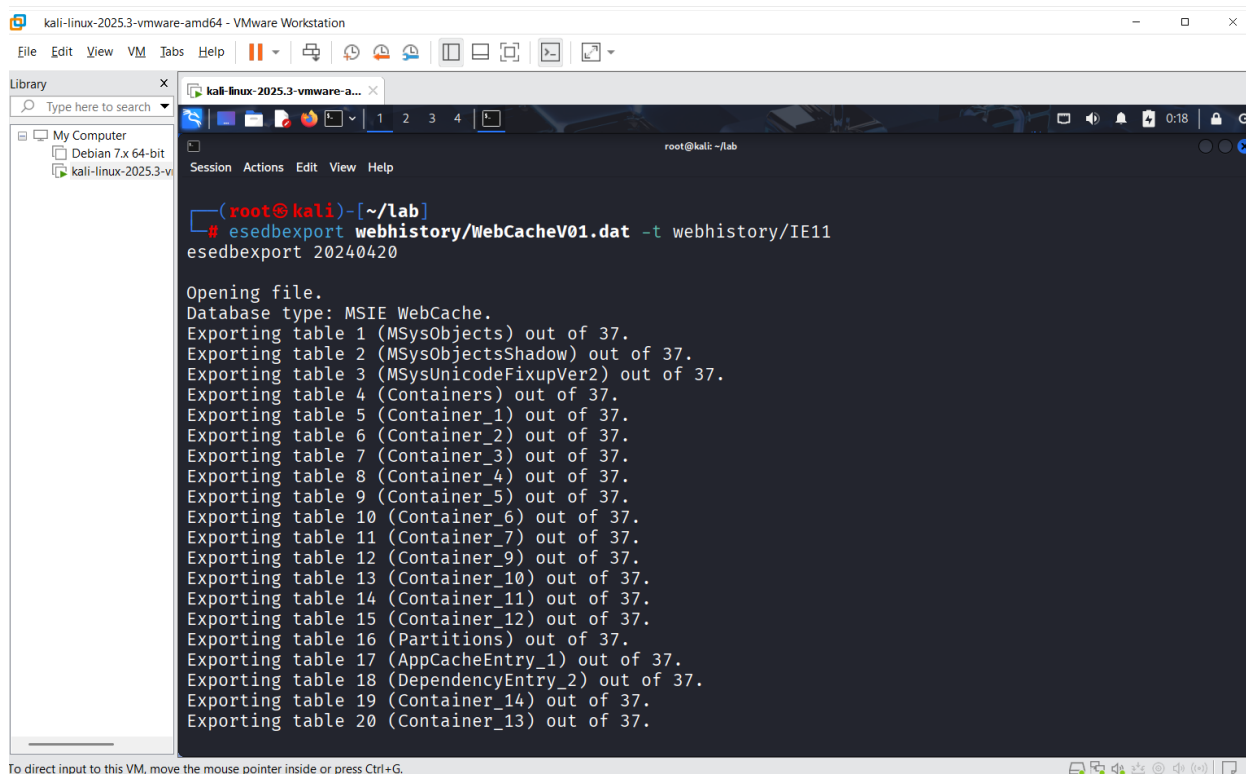
View IE 11 History Using *libesedb*



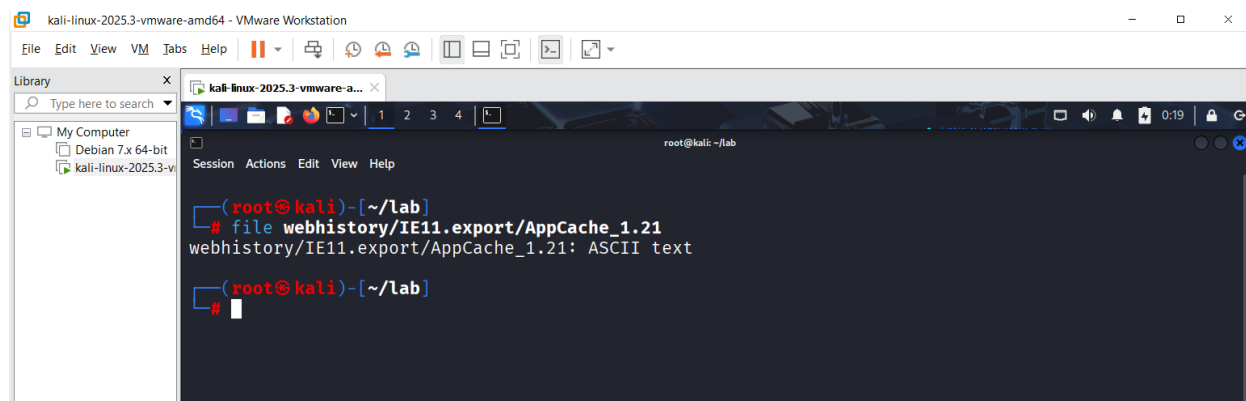
ảnh trên: Tìm loại tệp của WebCacheV01.dat



ảnh trên: Install *libesedb*



ảnh trên: Export *WebCacheV01.dat* to the folder *webhistory/IE11*



ảnh trên: Tìm loại tập tin ta thấy (ASCII text)

```
(root@kali)-[~/lab]
# echo -e "Frank manager 50000 \nAlex cleck 45000 \nEric clerk 25000"
Frank manager 50000
Alex cleck 45000
Eric clerk 25000

(root@kali)-[~/lab]
# echo -e "Frank manager 50000 \nAlex cleck 45000 \nEric clerk 25000" | awk '/manager/ {print}'
Frank manager 50000

(root@kali)-[~/lab]
# echo -e "Frank manager 50000 \nAlex cleck 45000 \nEric clerk 25000" | awk '{print $1, $3}'
Frank 50000
Alex 45000
Eric 25000

(root@kali)-[~/lab]
#

(kali-linux-2025.3-vmware-amd64 - VMware Workstation)
Applications
Session Actions Edit View Help

(root@kali)-[~/lab]
# echo -e "Frank manager 50000 \nAlex cleck 45000 \nEric clerk 25000" | awk '{print NR, $1}'
1 Frank
2 Alex
3 Eric

(root@kali)-[~/lab]
# echo -e "Frank manager 50000 \nAlex cleck 45000 \nEric clerk 25000" | awk 'sum +=3 ; END {print sum}'
9

(root@kali)-[~/lab]
#
```

2 ảnh trên: Ôn awk qua ví dụ 3 dòng “Frank manager 50000”

Ví dụ này dạy:

NR in số thứ tự dòng; chọn field \$1; tính tổng cột \$3.

```
(root@kali)~/lab
# cat webhistory/IE11.export/Container_1.4 | head -n 1
EntryId ContainerId CacheId UrlHash SecureDirectory FileSize Type Flags AccessCount SyncTime C
reationTime ExpiryTime ModifiedTime AccessedTime PostCheckTime SyncCount ExemptionDelta Ur
l Filename FileExtension RequestHeaders ResponseHeaders RedirectUrl Group ExtraData

(root@kali)~/lab
#

(root@kali)~/lab
# cat webhistory/IE11.export/Container_1.4 | head -n 1
EntryId ContainerId CacheId UrlHash SecureDirectory FileSize Type Flags AccessCount SyncTime C
reationTime ExpiryTime ModifiedTime AccessedTime PostCheckTime SyncCount ExemptionDelta Ur
l Filename FileExtension RequestHeaders ResponseHeaders RedirectUrl Group ExtraData

(root@kali)~/lab
#

(root@kali)~/lab
# awk '{print NR, $13, $18}' FS='\t' webhistory/IE11.export/Container_1.4 | head -n 5
1 ModifiedTime Url
2 Feb 27, 2015 00:47:45.000000000 https://technet.microsoft.com/favicon.ico
3 Mar 20, 2015 14:36:39.000000000 http://www.wired.com/wp-content/themes/Phoenix/assets/images/favicon.ico
4 Mar 20, 2015 14:36:39.000000000 http://www.wired.com/favicon.ico
5 Jan 01, 1601 00:00:00.000000000 https://license.piriform.com/verify?p=ccpro&c=cc&cv=5.04.5151&l=1033&l=CJ9T-J7
CU-SPNV-GWMB-WBEC&mk=YSNW-75UN-3WNW-QDJI-6UFT-T7TM-9AX9-8ZG8-USQZ

(root@kali)~/lab
#
```

2 ảnh trên: Hiển thị tên trường để biết thứ tự các trường, sau đó tách các trường bằng tab ‘\t’ và hiển thị ModifiedTime và URL để xem 5 dòng đầu. Ý nghĩa: kiểm tra nhanh dữ liệu đã “ra chữ” & biết chính xác cột nào chứa URL/timestamp.

```
(root@kali)~/lab
# awk '{print NR, $13, $18}' FS='\t' webhistory/IE11.export/Container_1.4 | wc -l
58

(root@kali)~/lab
# awk '{print NR, $13, $18}' FS='\t' webhistory/IE11.export/Container* | wc -l
10248

(root@kali)~/lab
#
```

ảnh trên: Đếm số bản ghi. Đầu tiên đếm trong một file Container_1.4, sau đó đếm tất cả file Container*

Step 3. Liệt kê keyword tìm kiếm (timestamp, URL, keyword)

từ URL truy vấn (query string), trích ra từ khóa người dùng gõ.

Mẫu URL tìm kiếm IE/Bing

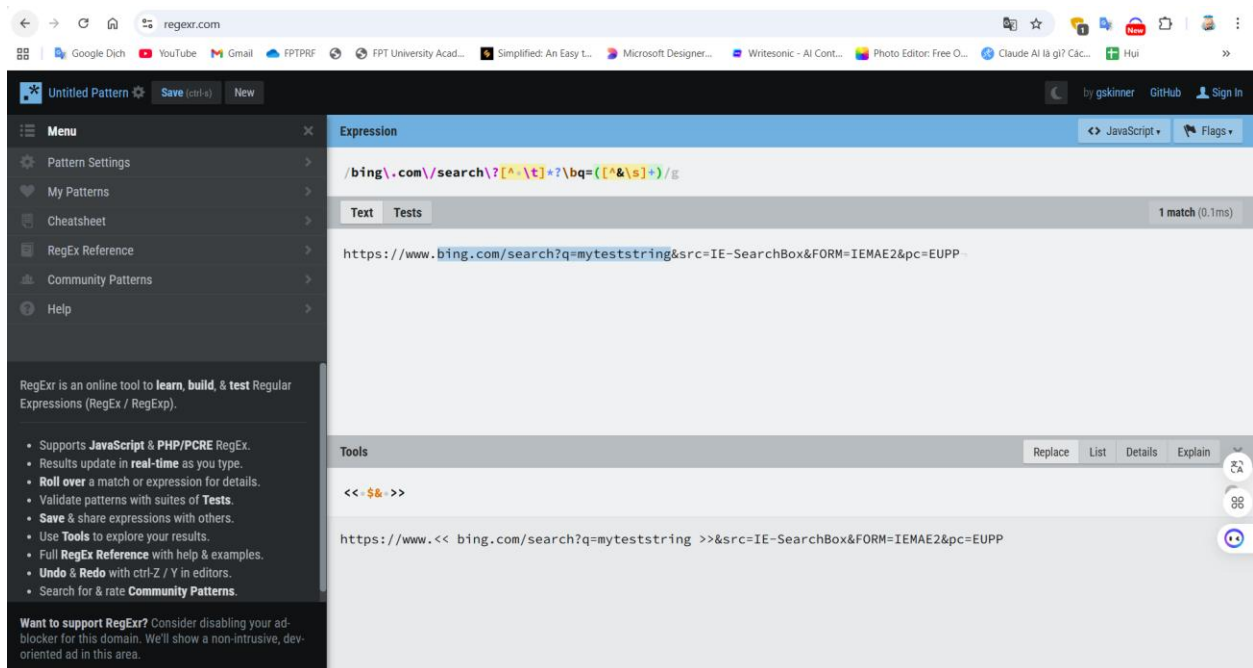
Ví dụ: <https://www.bing.com/search?q=myteststring&...> ⇒ tham số **q=** chính là keyword.

dùng regex để tự động nhận dạng URL tìm kiếm và bóc tách keyword, pattern hay dùng là `(bing\\.com\\search\\?[^\t]*?\bq=([^\s]+) :`

bing\\.com\\search\\? cố định domain + endpoint.

[^\t]*? không lấy khoảng trắng/tab giữa ? và tham số q=

\bq=([^\s]+) tìm đúng q=



Liệt kê IE 11 search keywords bằng awk:

kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Library

Type here to search

My Computer

Debian 7x 64-bit

kali-linux-2025.3-vm

kali-linux-2025.3-vmware-a...

root@kali: ~ /lab

Session Actions Edit View Help

```
(root@kali)-[~/lab]
# awk '{print $34}' webhistory/IE11.export/Container* | grep -P "bing\.com\/search\?q=(.*?)\&" --color
informant@http://www.bing.com/search?q=Top+Stories&FORM=HDRSC1
informant@http://www.bing.com/search?q=external%20device%20and%20forensics&qs=n&form=QBRE&ppq=external%20device%20a
nd%20forensics&sc=8-9&sp=-1&sk=&cv=c30c4b1f36114b1c9bc683838c69823a
informant@http://www.bing.com/search?q=anti-forensic+tools&qs=n&form=QBLH&ppq=anti-forensic+tools&sc=8-13&sp=-1&sk=
&cv=e799e715fa2244a5a7967675bdcca9d3
informant@http://www.bing.com/search?q=file+sharing+and+tethering&qs=n&form=QBLH&ppq=file+sharing+and+tethering&sc=
0-18&sp=-1&sk=&cv=171b77e4ffd54b2a92c4e97abf995fe1
informant@http://www.bing.com/search?q=DLP%20DRM&qs=n&form=QBRE&ppq=dlp%20drm&sc=8-7&sp=-1&sk=&cv=6e206ee8751e4ad
89f882ed52daf3aea&sid=BE5E388F8757406CAA32E58334719A20&format=jsonv2&jsoncbid=0
informant@http://www.bing.com/search?q=what%20is%20windows%20system%20artifacts&qs=n&form=QBRE&ppq=what%20is%20wind
ows%20system%20artifacts&sc=0-27&sp=-1&sk=&cv=1ef4ace146854d97acf263b53bf97b8c&sid=BE5E388F8757406CAA32E58334719
A20&format=jsonv2&jsoncbid=4
informant@http://www.bing.com/search?q=Top+Stories&FORM=HDRSC1
informant@http://www.bing.com/search?q=external%20device%20and%20forensics&qs=n&form=QBRE&ppq=external%20device%20a
nd%20forensics&sc=8-9&sp=-1&sk=&cv=c30c4b1f36114b1c9bc683838c69823a
informant@http://www.bing.com/search?q=external%20device%20and%20forensics&qs=n&form=QBRE&ppq=external%20device%20a
nd%20forensics&sc=8-9&sp=-1&sk=&cv=c30c4b1f36114b1c9bc683838c69823a&sid=BE5E388F8757406CAA32E58334719A20&format=
jsonv2&jsoncbid=9
informant@http://www.bing.com/search?q=Forensic+Email+Investigation&FORM=QSRE1&sid=BE5E388F8757406CAA32E58334719A2
0&format=jsonv2&jsoncbid=3
informant@http://www.bing.com/search?q=cd%20burning%20method&qs=n&form=QBRE&ppq=cd%20burning%20method&sc=8-2&sp=-1&
sk=&cv=b7dbe6fb67424c578172ba57330a0894&sid=BE5E388F8757406CAA32E58334719A20&format=jsonv2&jsoncbid=7
informant@http://www.bing.com/search?q=e-mail+investigation&qs=n&ppq=e-mail+investigation&sc=8-7&sp=-1&sk=&cv=fel
c3738d8c7471284731724166959af&first=9&FORM=PERE&sid=BE5E388F8757406CAA32E58334719A20&format=jsonv2&jsoncbid=2
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.